Homework 2 Question 4

a) 3C2

Binary:

3 C 2

0011 1100 0010

convert to 4 binary digits

$= \boxed{(001111000010)_2}$

Octal:

3C2

0011 1100 0010

First convert to 4 binary

001 111 000 010

Then convert to 3 binary

$\boxed{(1702)_8}$

Decimal:

$3C2_{16} = 3 \times 16^2 + 12 \times 16^1 + 2 \times 16^0$

convert letter to decimal C=12

multiply by base 16 and decimal place.

$= \boxed{(962)_{10}}$

b)

To compute the sum and product, convert both to decimal and then convert back to hexadecimal after calculation

1) 1A01 to decimal

$1 \cdot 16^3 + 10 \cdot 16^2 + 0 \cdot 16^1 + 1 \cdot 16^0$

$= \boxed{6657}$

2) find sum

$6657 + 8020 = \boxed{14677}$

3) find product

$6657 \times 8020 = \boxed{53{,}389{,}140}$

4) convert to hex:

14677 / 16 = 917 R 5

917 / 16 = 57 R 5

57 / 16 = 3 R 9

3 / 16 = 0 R 3

$= \boxed{(3955)_{16}}$

53,389,140 / 16 = 3336821 R 4

3336821 / 16 = 208551 R 5

208551 / 16 = 13034 R 7

13034 / 16 = 814 R 10

814 / 16 = 50 R 14

50 / 16 = 3 R 2

3 / 16 = 0 R 3

$= \boxed{(32EA754)_{16}}$